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TechSprint

Leveraging the power of AI

Team Details

a. **Team name: CoreTech Labs**
b. **Team leader name: Yash Dhadge**
c. **Problem Statement: Open Innovation**

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Gamified Learning Platform for Education

Proposed Solution

- Mobile or Web Application containing interactive games Grade Wise,Subject wise and Topic Wise
- Students will select the subject / topic and play related games
- They are then creditted points based on the correctly answered questions
- Ranking based on points and leaderboard
- Teacher Dashboard with student analytics and performance Offline access for areas with limited internet access

Our Model Promises:

- Provides: Grade,subject,and topic-wise interactive games for targeted practice
- Ensures Offline-first learning flows and Progress Tracking
- Supports :Teacher Dashboards and Student engagement through rewards, points and badges
- Improves: Motivation and Retention and improve visibility of progress through leaderboards

Prototype Video :https://youtu.be/4vRblgKul40

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OPPORTUNITIES: Market & Differentiation

Apps in Market

- Generic curriculum (not localized)
- Requires consistent internet
- Limited language support
- Complex dashboards
- Cloud-dependent (offline not supported)
- No gamification elements

Our app

How Different?

- Offline-first architecture (preloaded lessons, batch sync)
- Low-end device optimized (runs on basic phones)
- Multilingual + voice hints (local language support
- Rural-focused curriculum (Grade 6-12, topic-wise)
- Teacher dashboards (simple performance tracking)

Problem Solving Approach

- Offline + sync = tackles intermittent connectivity
- Gamification (badges, points, leaderboards) = +15% engagement
- Simple dashboards = teacher empowerment
- Mobile + Web = device flexibility
- Local languages = universal accessibility

Business Opportunity

- **CAGR: 33.6% (fastest-growing ed-tech)**
- **37% boost in progress tracking with gamification**
- **60% rural India = underserved market**

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List of features offered by the solution

1.Admin Dash Board

Manage institute

Manage User

Approvals

Global Analytics

Feedback Review System

2.School Login

Mange Teacher/Students

Syllabus/Topic

Set Performance Measures

3.Teacher Dash Board

Teaching Plan

Quiz formation

Students Performance Report

Student Progression

Slow and Advance Learner

Assigning Collaborative Learning

4. Student Dash Board

Understanding Subject Concepts

Learning through Games

Earn Badges

Attempt Quiz

5. Digital Wellbeing Monitoring System for students

6. Mapping of Institutes, Teachers and Students on Gmap.

7. Notifications through App

8. Compatibility

9. Works Offline for remote area

10. ML model for Course Outcome and Learning Outcome

11. Slow and Advance Learner

12. Teacher Feedback

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Google Technologies used in the solution

To visualize the schools/institutes, students, teachers from various locations

The AI Tutor/ChatBot is powered by gemini and is also used by teachers to create various tests

The Application is to be hosted on Google Cloud and also uses Google authentication

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Wireframes/Mock diagrams of the proposed solution (optional)

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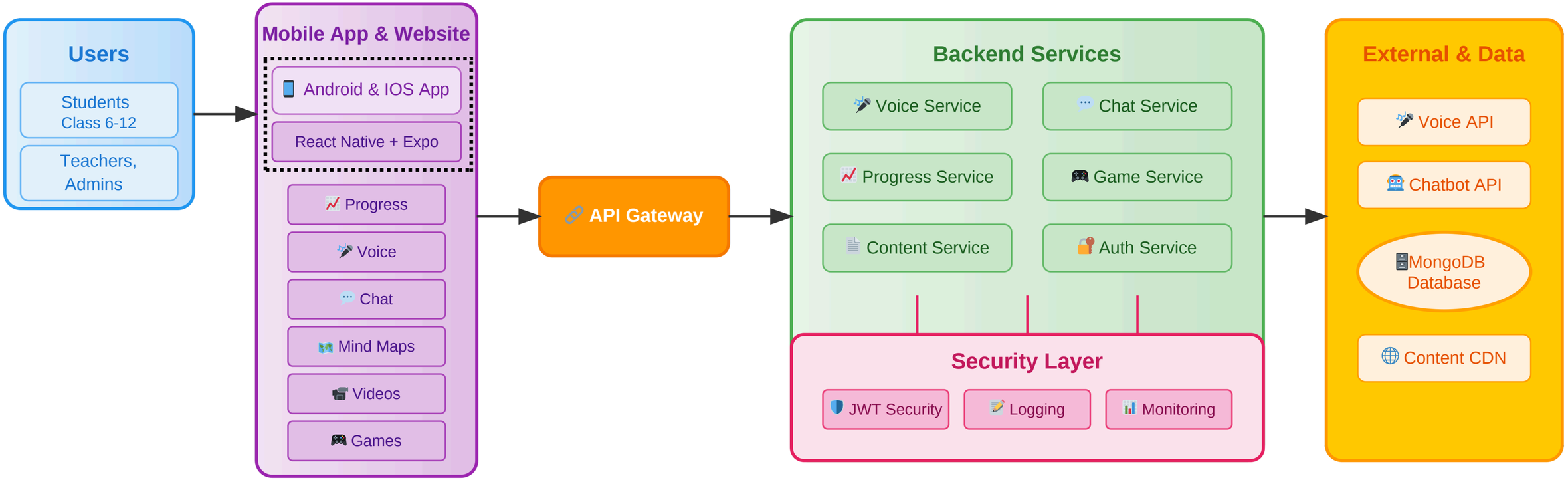
Architecture diagram of the proposed solution



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Future Scope & Roadmap

Current Limitations:

- Multi-board standardization for CBSE, State Board, and ICSC
- Internet connectivity requirement for AI chatbot

Future Development Plans:

- Multi-language support for regional boards
- Advanced AI analytics for personalized learning
- Integration with school management systems
- Feature similar to NoteBookLM which generates PPT, flashcards, infographics, mindmaps, animated videos, podcasts, and many more...

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Provide links to your:

1. **GitHub Public Repository - Github**
2. **Demo Video Link (3 Minutes) - https://youtu.be/4vRblgKul40**

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